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Faculty of Medicine of Constantine

Cytology Courses

THE CYTOSKELETON

I. Introduction

The cytoskeleton constitutes a dynamic framework at the subcellular scale. In eukaryotes, it is composed of networks of fibrous polymers (microfilaments, microtubules, and intermediate filaments) associated with numerous accessory and motor proteins. These polymers are classified into three main types based on their diameter:

- Actin microfilaments (MF), with a diameter of approximately 7 nm.
- Intermediate filaments (IF), with a diameter of approximately 10 nm.
- Microtubules (MT), with a diameter of approximately 25 nm

II-Structural elements of the cytoskeleton:

II.1. Actin microfilaments:

These are flexible, polarized filaments with a diameter of approximately 7 nm. They are formed through the polymerization of globular actin monomers (G-actin), which assemble into a double-stranded helical structure known as filamentous actin (F-actin). (**Figure 1**)"

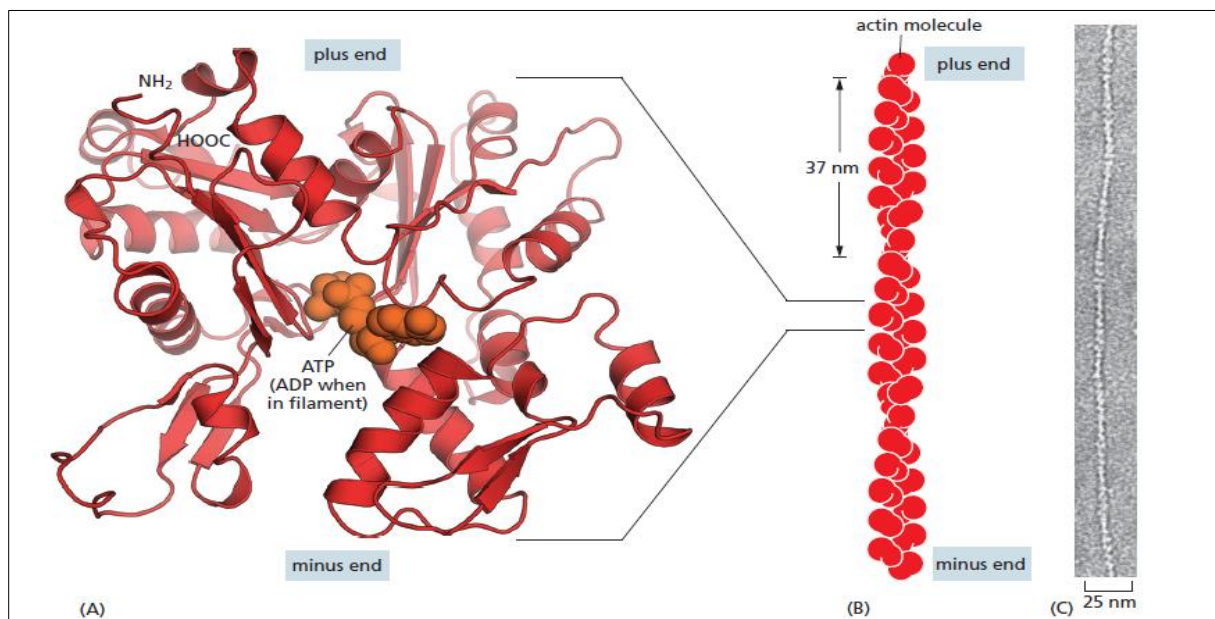


Figure 1 : Tertiary structure of G-actin (A). Fibrillar actin or F-actin (B). Electron micrograph of an actin microfilament (C).

II.1.1. Polymerization of G-actin into F-actin: In addition to ATP, the polymerization of G-actin into F-actin requires divalent cations, specifically magnesium (Mg^{2+}). These ions act as essential cofactors that stabilize the monomeric conformation and facilitate its incorporation into the growing filament (**Figure 2**).

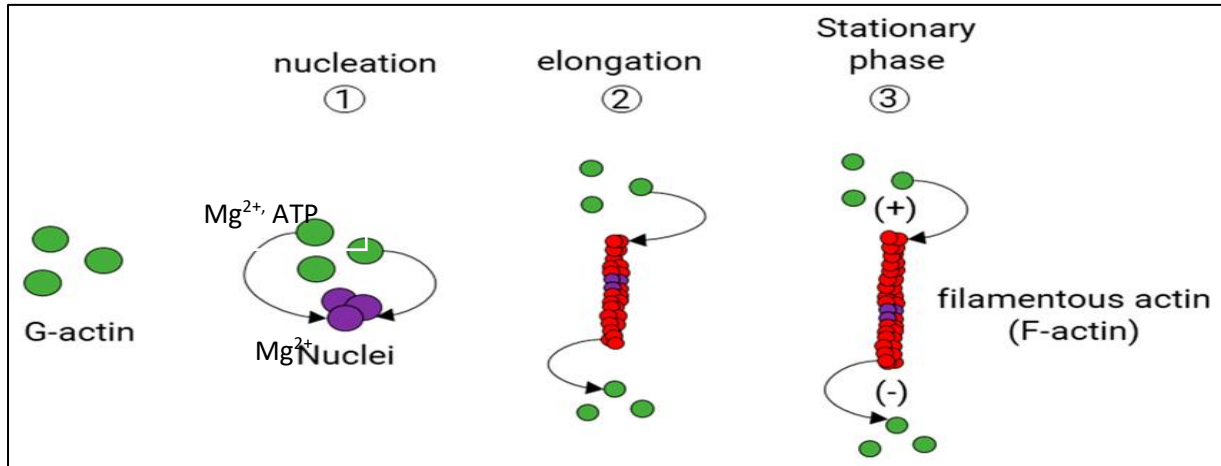


Figure 2 : Polymerization of G-Actin (globular) into F-actin (Filamentous).

F-actin has two ends:

- The Plus (+) End or "Barbed" End: This is the fast-growing filament end.
- The Minus (-) End or "Pointed" End: This is the pole of slow growth (or rapid depolymerization).

II.1.2. Proteins associated with actin microfilaments: The dynamics of microfilaments are regulated by a wide variety of actin-Binding proteins (ABPs) (**Table 1**).

II.1.3. Function of actin microfilaments (Figure 3): Actin microfilaments perform a multitude of essential functions within the cell:

- They participate in the movement of organelles and vesicles within the cell, through association with **Myosin V** and **Myosin VI** (**Figure 3. A**)
- They are involved in muscle contraction (**Figure 3.B**).
- They constitute the cell cortex, which is involved in cell mobility and the movements of the plasma membrane (formation of lamellipodia, filopodia, and pseudopodia) (**Figure 3.C**).
- They form the contractile ring in dividing cells (**Figure 3.D**).
- They form the structural core of intestinal microvilli (**Figure 3.E**).

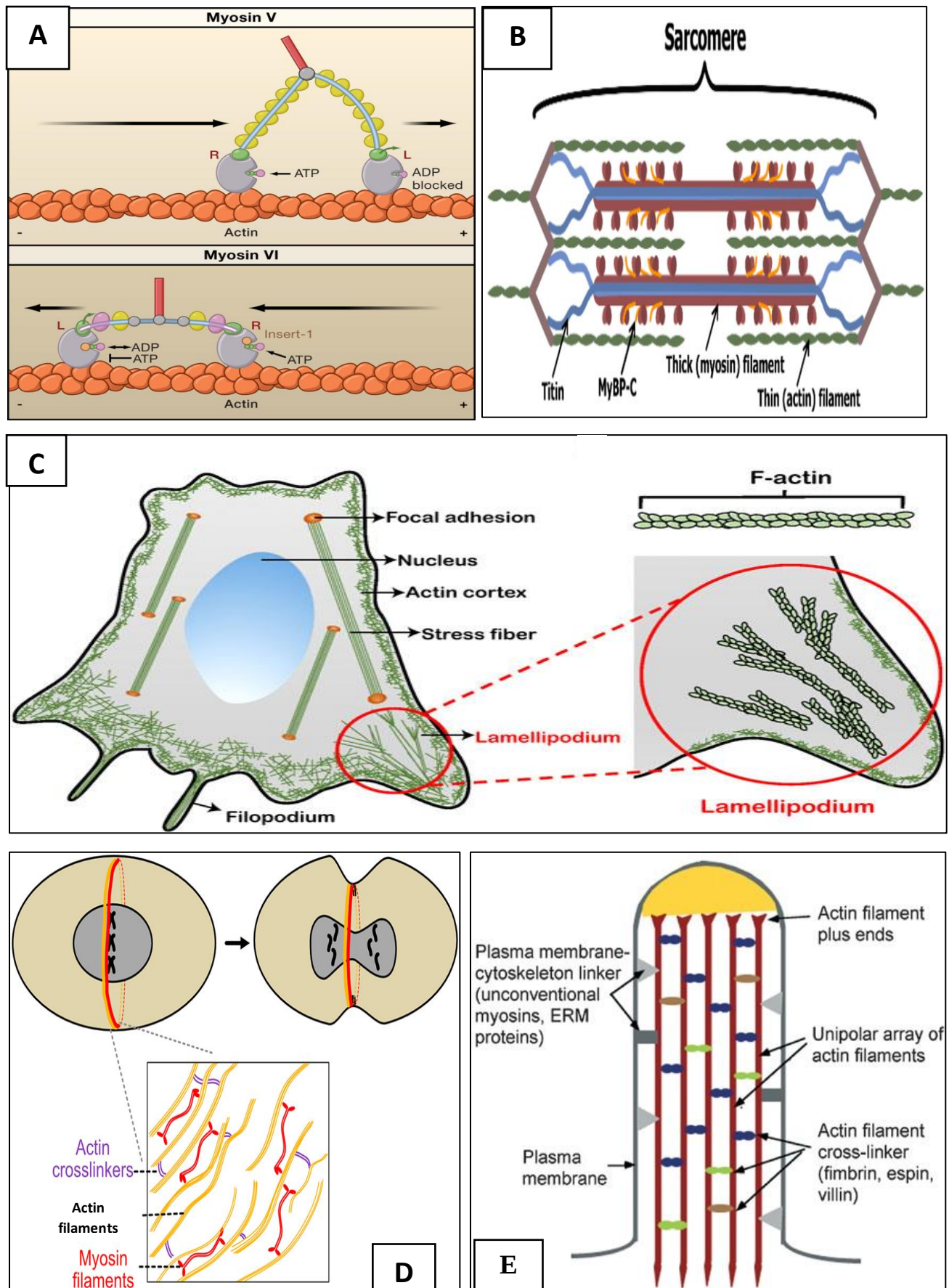


Figure 3: The various functions performed by actin microfilaments in human cells.

Table I: Actin Microfilament-Associated Proteins. *Schematized on the slides*

Category	Mechanism of Action	Examples
1. Sequestering Proteins	Inhibit polymerization by binding G-actin monomers.	Thymosin β4
2. Polymerization-Promoting Proteins	Accelerate polymerization at the plus (+) end.	Profilin
3. Nucleating Proteins	Initiate the formation (nucleation) of new filaments.	Arp2/3 complex: forms branched networks (70° angle). Formins: promote long, straight (unbranched) filaments.
4. Capping Proteins	Stabilize filaments by preventing polymerization or depolymerization at the ends.	CapZ (plus end), Tropomodulin (minus end)
5. Severing / Fragmentation Proteins	Break filaments into shorter fragments to promote disassembly or new growth.	Gelsolin, Cofilin
6. Cross-linking Proteins	Organize filaments into bundles or 3D networks.	α-actinin, Fimbrin (bundles); Filamin (networks)
7. Motor Proteins	Utilize ATP to drive contraction and intracellular transport.	Myosin II (contraction); Myosins V & VI (vesicle trafficking)
8. Membrane-Binding Proteins	Anchor actin filaments to transmembrane proteins or phospholipids.	Dystrophin, Spectrin, ERM family (Ezrin, Radixin, Moesin)

II.2. Microtubules

II.2.1. Organization and Ultrastructure of Microtubules: Microtubules are hollow, rigid cylinders resulting from the polymerization of a globular protein: tubulin.

- **The Structural Unit:** The basic building block is a heterodimer composed of two closely linked subunits: α -tubulin and β -tubulin. Each subunit binds a GTP molecule (**Figure 4**).
- **Biochemical Detail:** While α -tubulin keeps its GTP intact, β -tubulin possesses GTPase activity, allowing it to hydrolyze GTP into GDP once the dimer is incorporated into the filament (**Figure 6**).
- **Formation of Protofilaments:** Dimers assemble longitudinally to form protofilaments. This organization confers both structural and functional polarity to the microtubule:

a "plus" (+) end directed toward the cell periphery (where β -tubulin is exposed) and a "minus" (–) end directed toward the cell center (where α -tubulin is exposed).

- **Microtubule Architecture:** Thirteen protofilaments associate laterally to form the wall of a hollow cylinder, approximately 25 nm in diameter.
- **Helical Organization:** The arrangement of these protofilaments is not perfectly linear; they follow a helical organization, creating a slight structural offset known as the "seam" where the protofilaments do not perfectly align (**Figure 5**).

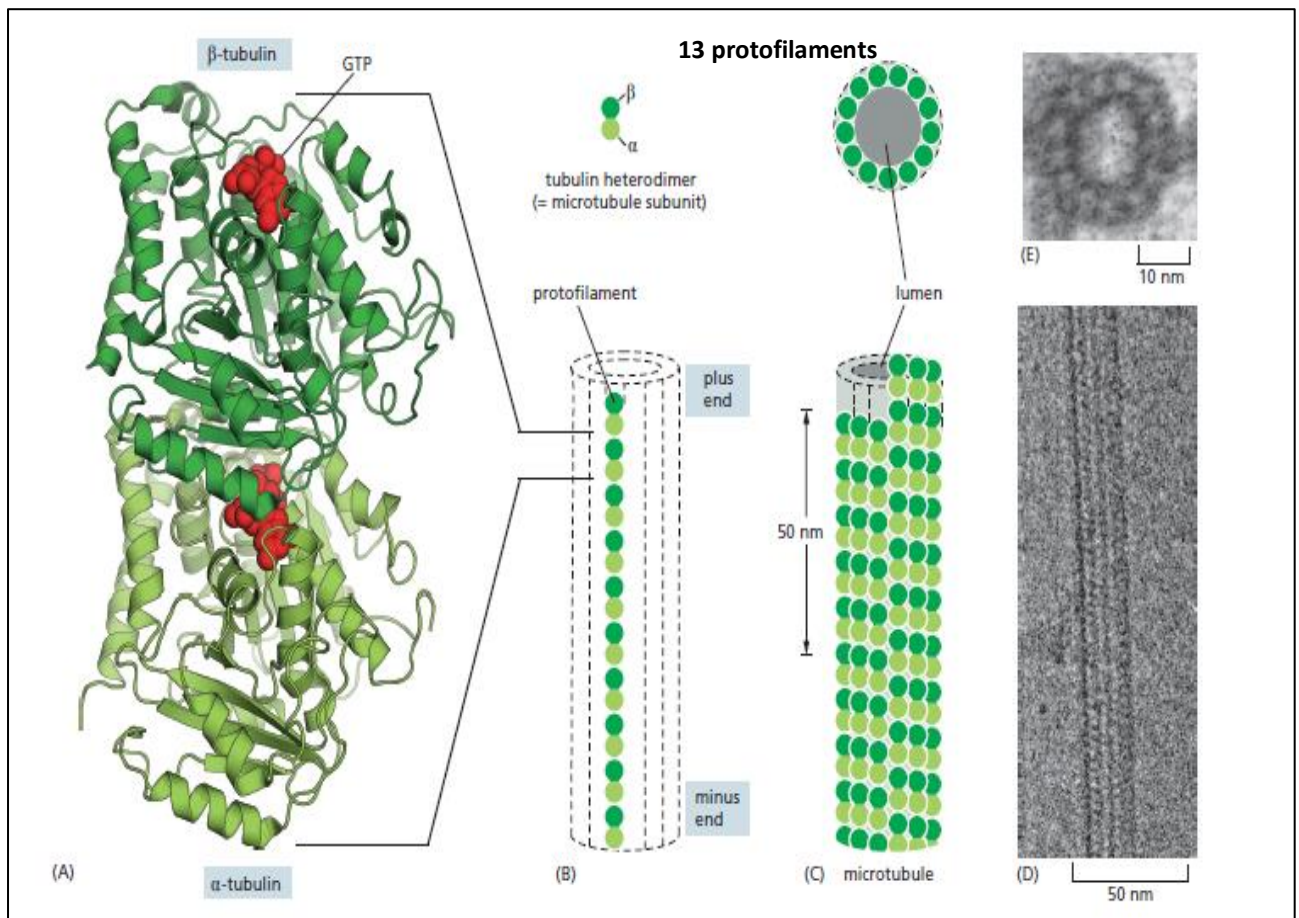


Figure 4: Tertiary structure of $\alpha\beta$ tubulin dimer (A). Arrangement of α - β dimers into protofilaments (B). Cross-sectional view of a microtubule: hollow cylinder made up of 13 tubulin protofilaments (C). Electron micrograph of a microtubule (D).

II.2.2. Microtubule-associated proteins (MAPs): Two main categories of MAP are distinguished (**figure 7**).

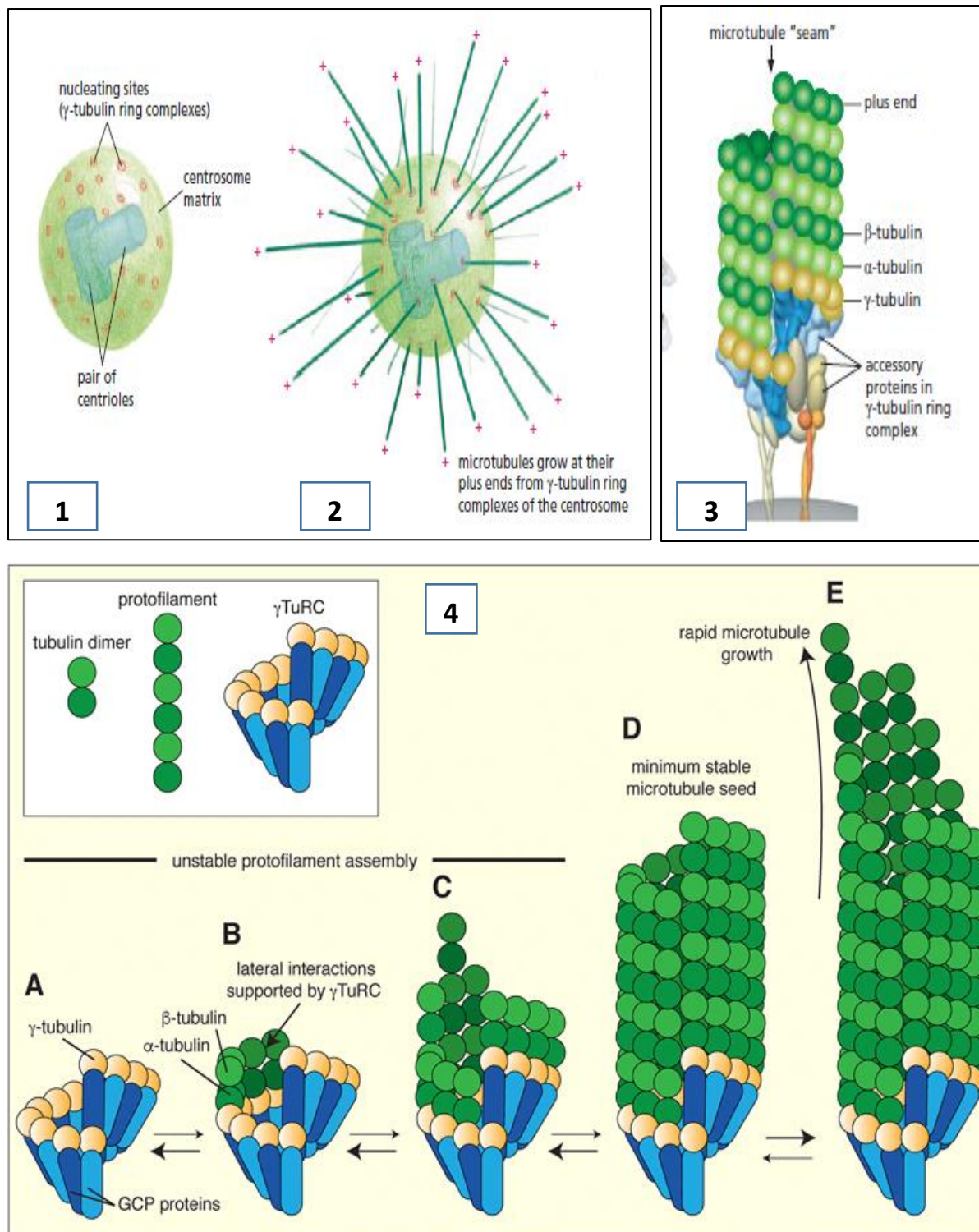


Figure 5: Within the cell, microtubule nucleation does not occur spontaneously, but begins at a Microtubule Organizing Center (MTOC), such as the centrosome (1 and 2). This process requires a specific protein complex: the γ -TuRC (Gamma-Tubulin Ring Complex). Arranged in a ring, this complex serves as a template (or "mold") onto which tubulin heterodimers assemble. By "capping" the minus (-) end, the γ -TuRC not only defines the number of protofilaments (generally 13), but also ensures the stability of this end, leaving the plus (+) end free to grow towards the cell periphery (3 and 4).

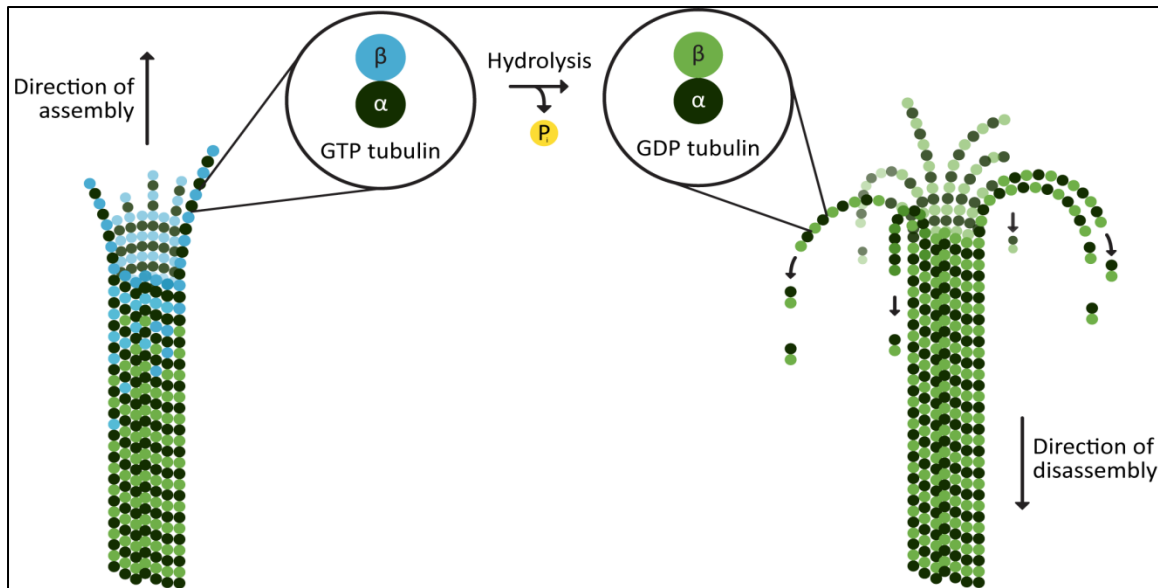


Figure 6: Polymerization and depolymerization of microtubules.

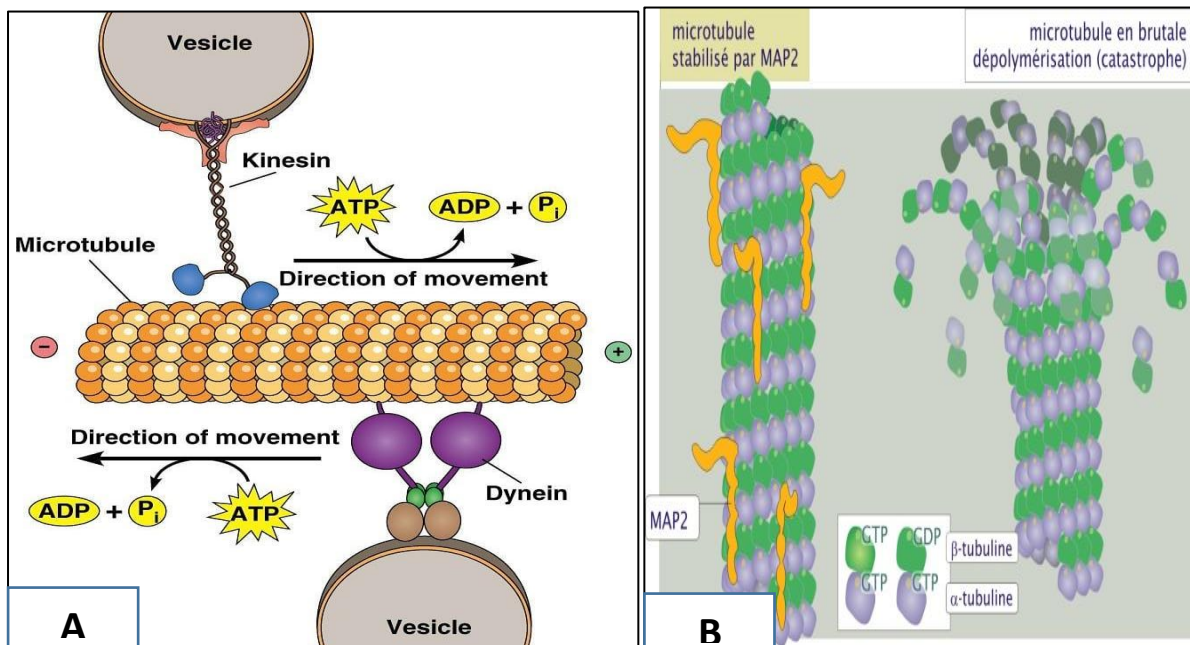


Figure 7: The motor MAPs (molecular motors) (A). Structural MAPs (B)..

- (a) Structural MAPs (Stabilizing Proteins):** These proteins bind along microtubules to regulate their stability. Proteins like the protein Tau (abundant in axons) or MAP2 (in the dendrites) and MAP4 in the rest of the cells; stabilize the filaments against depolymerization and determine the spacing between microtubules, thus organizing the architecture of the cytoskeleton. b)
- (b) Motor MAPs (Molecular Motors):** These proteins use ATP to move along microtubules. They ensure the transport of vesicles and organelles. We distinguish:

Kinesines, which generally carry out anterograde transport (towards the end+), and **Dyneins**, which ensure retrograde transport (towards the end-).

II.2.3. Functions of microtubules: Microtubules perform several functions within the cell:

- They participate in the intracellular transport of vesicles and organelles.
- They are part of the organization of centrioles, cilia, and flagella (stables MT).
- They participate in cell division through the formation of the mitotic spindle.
- They contribute to the cellular polarity and to the orientation of movements related to membrane flow.

II.2.4. Stable (non-dynamic) microtubules: These are long-lived structures with a stability nearly equivalent to the lifespan of the cell. They are protected against depolymerization by specialized stabilizing MAPs (Microtubule-Associated Proteins). These stable microtubules are primary structural components of cilia, flagella, and centrioles (**Figures 8 and 9**).

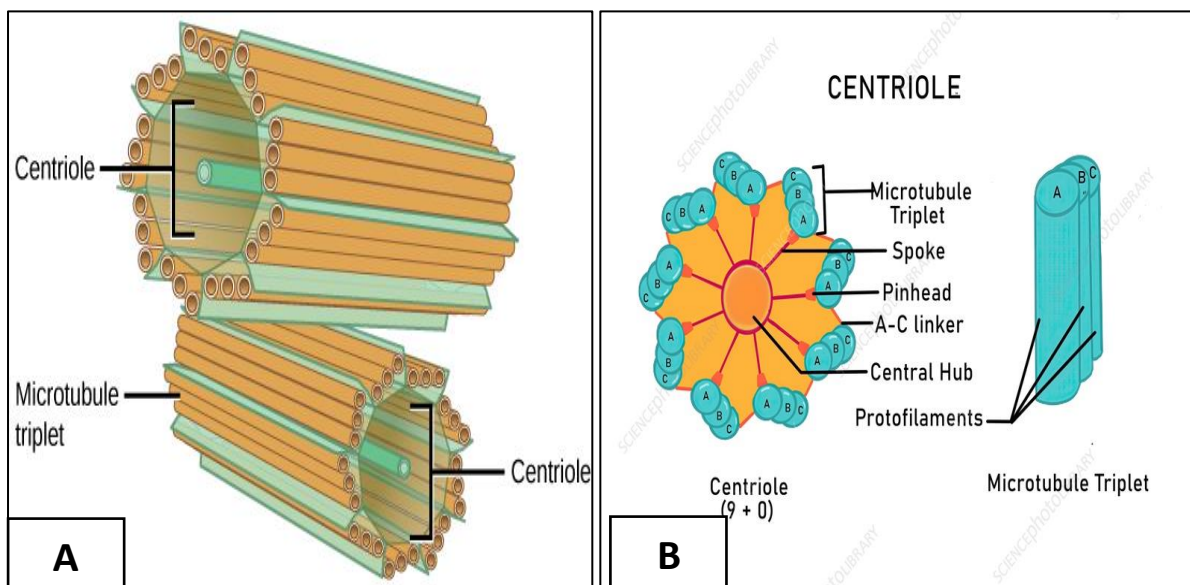


Figure 8: General organization of the centrosome (A). Transverse section of a centriole, showing the characteristic 9+0 arrangement of microtubule triplets (B).

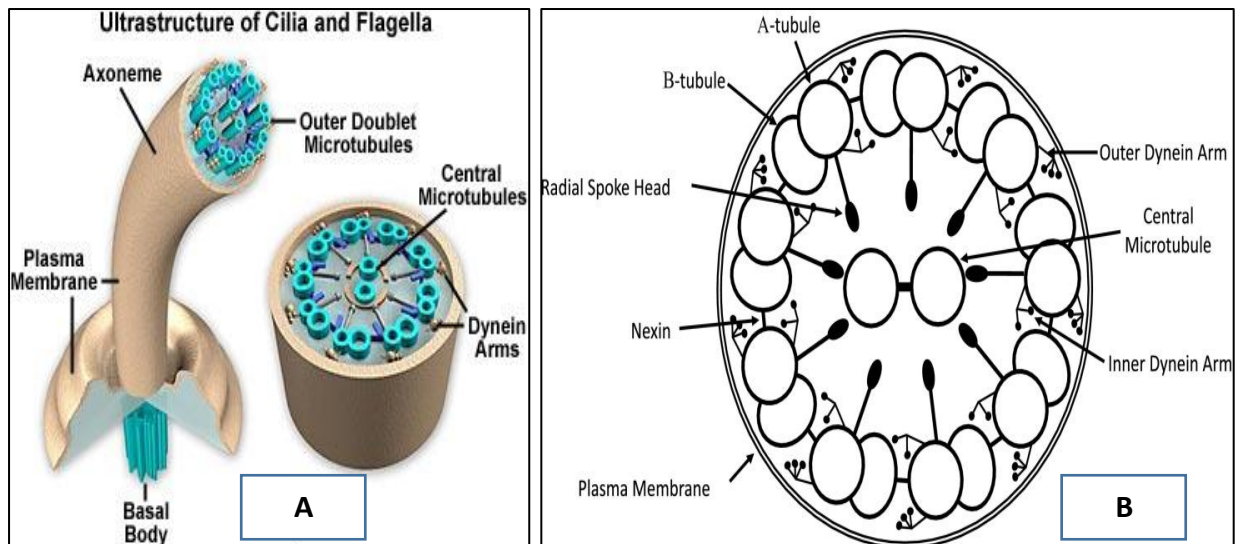


Figure 9. Transverse section of cilium and flagellum (A). Axonéme structure (9 doublets of microtubules at the periphery and 2 microtubules in the center; axoneme in a 9+2 arrangement) (B).

II.3. Intermediate Filaments (IF):

In contrast to actin microfilaments and microtubules, intermediate filaments are composed of fibrous rather than globular protein subunits. With a diameter of approximately 10 nm, they occupy an intermediate size between the other two cytoskeletal components. These filaments are notably apolar (lacking (+) and (–) ends), a characteristic that contributes to their role as the most stable and mechanically resilient elements of the cytoskeleton.

In the cytosol: Intermediate Filaments:

- Constitute rigid fibrous networks around the nuclear envelope and form a perinuclear cage.
- Interact with the plasma membrane at specific cell junctions (Desmosomes and hemidesmosomes).
- Reinforce cells against mechanical stress.

In the nucleus: A specific class of IF (Intermediate Filaments) called "lamins" forms a network lining the inner nuclear membrane on its nucleoplasmic face.

II.2.1. Organization of intermediate filaments: IF are formed by the polymerization of fibrous proteins (monomers). Their assembly occurs in several stages (**Figure 10**). In cross-section, an IF is composed of 32 monomers.

The main protein families grouped within the intermediate filament (IF) superfamily.

Type I IF: Cytokeratin: all epithelial cells.

Type II IF: Vimentin: in mesenchymal cells. Desmin: in muscle cells.

Type III IF: Neurofilaments: neurons (central and peripheral nervous system).

Type IV IF: Lamins A, B and C: found in the nucleus of all animal cells.

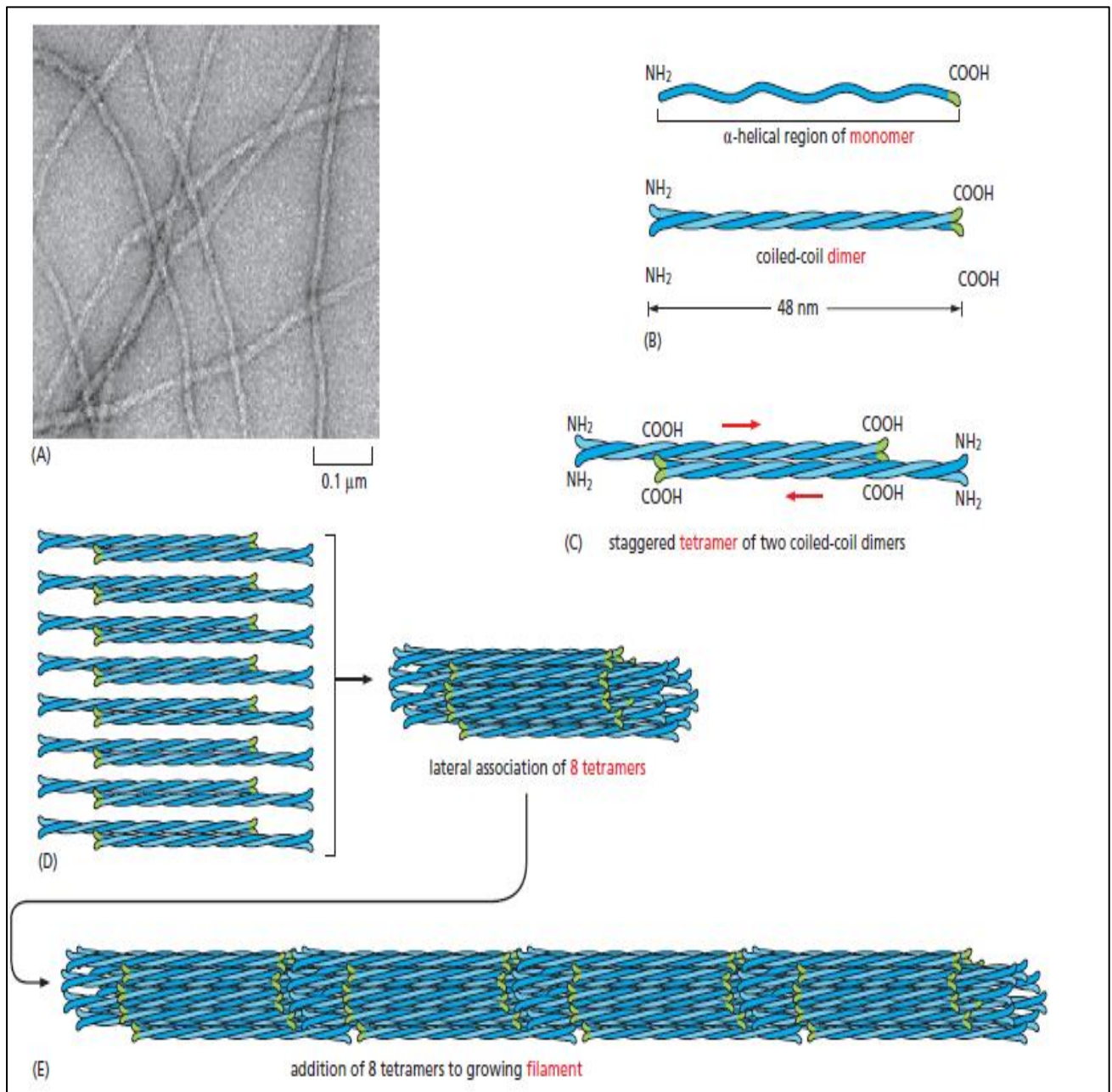


Figure 10 : Ultrastructure and organization of intermediate filaments.

III. Examples of pathologies associated with cytoskeleton dysfunction:

III.1. Non-muscle actin microfilament dysfunction:

Wiskott-Aldrich Syndrome is a genetic immunodeficiency caused by the absence of the regulatory protein WASP (Wiskott-Aldrich Syndrome Protein). This deficiency prevents the activation of the **Arp2/3 complex**, thereby blocking the nucleation and branching of actin microfilaments. This results in the inability of leukocytes to move and form immunological synapses.

III.2. Microtubule dysfunction:

Neurodegenerative diseases (Tauopathies): The most well-known is **Alzheimer's** disease. The Tau protein, which normally stabilizes microtubules in neurons, becomes hyperphosphorylated and detaches from them. Deprived of their "glue," the microtubules disassemble (depolymerize) and lose their functions. Additionally, the Tau protein clusters to form insoluble aggregates (neurofibrillary tangles) within the cytoplasm, which disrupt organelle function (such as mitochondria) and ultimately trigger apoptosis (programmed cell death).

III.3. Intermediate filament dysfunction:

Progeria (or Hutchinson-Gilford Syndrome) is a rare genetic laminopathy caused by the production of **progerin**, a mutated form of **lamin A**. This causes deformation of the nuclear envelope and the cell's inability to divide properly.